

Controlling pests

Pests are animals that impede human activities. They tend to be able to live in a wide variety of habitats, and can damage crops, spread disease, or infest homes. Pest control usually involves the use of chemicals, but this can cause pollution. By contrast, biological control makes use of natural predators and parasites to control pests.

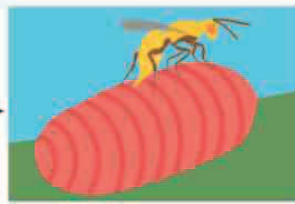
▽ Fly pests

Flies that lay their eggs in animal waste and rotting food are a serious pest in stables, farms, and sewage treatment plants. Their maggot larvae eat the waste before pupating into adults that can spread diseases.



▽ Parasitic wasp

The spalangia wasp is native to Australia, but humans use it to kill flies naturally around the world. The wasp lays its eggs inside a fly pupa. The wasp larva hatches and eats the fly pupa from the inside out.



▽ Becoming an adult

The wasp then pupates itself inside the empty fly case. When fully grown, the adult wasp bites a hole in the case and flies away. After mating, the female wasps will lay more eggs on fly pupae, until all the flies are dead.



Introduced species

As humans move around the world, they take animals and plants with them. Introducing species to new habitats can upset the balance of an ecosystem. There have been several disastrous introductions, such as the introduction of the cane toad to Australia (below), or the 60 starlings that were released in New York City, in 1890. There are now 200 million starlings in North America, which have pushed many native birds close to extinction.

Key
 Spread of cane toads today



△ Beetle pest

The grubs of a native beetle were damaging valuable plantations of sugar cane across tropical parts of eastern Australia. Farmers looked for a small predator to control the beetle numbers and protect the crops.



△ Marine toad

A large toad from South America was introduced in 1936 to tackle the beetles. It was known as the marine toad because it was tough enough to survive almost anywhere, even along the seashore.



△ Spread across Australia

The toads ate almost everything except the cane beetle, upsetting delicate ecosystems. They spread and became a major pest, renamed the cane toad. There are more than 200 million living in Australia today.

REAL WORLD

Genetic modification

Humans have been altering the genetic makeup of animals and plants for thousands of years by selectively breeding animals with desired characteristics. However, in recent years, genetic engineers have been adding completely new genes to animals in the laboratory. These fish glow in the dark because of a gene added from a bioluminescent (light-emitting) deep sea jellyfish.



Humans are the only species to live on all **seven continents**, including a permanent settlement in the Antarctic since 1956.